

Question	Answer	Mark	Mark scheme	Additional guidance
18 (a)	200	P1	for $\frac{55}{11}$ (= 5) or $\frac{11}{55}$ (= 0.2) or $\frac{40}{11}$ (= 3.63..) or $\frac{11}{40}$ (= 0.275) or for starting to use ratio eg tagged : untagged = 11 : 44 oe	Allow decimal values truncated or rounded to at least 2dp
		P1	for $\frac{11}{55} = \frac{40}{n}$ or $\frac{11}{40} = \frac{55}{n}$ or $\frac{55 \times 40}{11}$ oe or for process to find the number of untagged frogs eg $\frac{55-11}{11} \times 40$ (= 160)	
(b)	Assumption	A1	cao	May be seen in ratio eg 40 : 160 A correct answer with no supportive working gets 0 marks.
		C1	for a suitable assumption, Acceptable examples no frogs have left/arrived at the lake no tags have fallen off / no lost tags sample of frogs taken was a random sample / this is a representative sample tagged frogs are distributed around the lake no baby frogs / no frogs have died There are (still) 4 untagged for every tagged frog. The tagged frogs had time to blend into the population. It is a closed environment There aren't any more frogs in the lake No other frogs in the lake Not acceptable examples She has caught all the frogs There are lots of frogs. This is the same lake	